بسم الله الرحمن الرحيم

CS458 - Data Mining (DM)

Fall 2025

Lecture 1: Introduction

Rafia Asad Khan, University of Management and Technology, Lahore

Introduction

Instructor, Students and Course

Instructor

- Rafia Asad Khan
  - Lecturer
    - Office: SST1-708
    - Counseling Hours:
      - Monday – Friday (9:00 am – 11:00 am),
    - Email: rafia.asad@umt.edu.pk

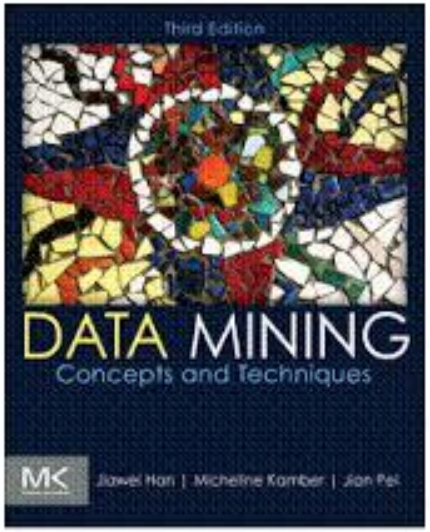
Course Objectives

- Objectives
  - To develop the concepts of and the techniques in key data mining tasks
  - To provide hands-on experience with data mining using tools
  - To encourage innovative and useful applications of data mining tasks
  - Explore, visualize, and analyze large datasets
  - Select and evaluate data mining techniques for the discovery of relevant knowledge from datasets
  - Understand efficiency, scalability, and correctness challenges in data mining

Textbook(s)/Supplementary Readings

- Data Mining: Concepts and Techniques,
  - J. Han, M. Kamber, and J. Pei,
  - Third Edition, Morgan Kaufmann Publishers, 2012.

  Reference:
- Introduction to Data Mining,
  - V. Tan et al.
  - Addison-Wesley, 2009.
- Data Mining: Practical Machine Learning Tools and Techniques,
  - Ian H. Witten, Eibe Frank and Mark A. Hall,
  - Third Edition, Morgan Kaufmann Publishers, 2011



- Tools and Technologies
  - Weka
  - C++ or Java, Matlab, Python

Grading Policy

Instrument	Description	Weight
Class Exercises	In-class exercises and evaluation	10%
Assignments	Assigned during important stages of the course to apply and practice the learnt concepts	5%
Project	One group project	10%
Quizzes	In-class (un)announced 15 minutes tests	15%
Mid-Term Exam	A single 90-minutes exam from the material covered during the first 6-7 weeks	20%
Final Exam	Will cover the entire course. At least 70% of the material would be post mid term.	40%

Late Submission Policy: Late submissions are not allowed

Classroom Policy

1. Attendance is very important, 80% is required, 100% is recommended.
2. Keep your mobiles switched off
3. Females sit on the right while facing white board
4. Quizzes can be announced or unannounced. 1 quiz would be dropped out of 4 or 5 quizzes. No retake for quizzes.
5. The plagiarism and cheating cases would be reported to the Disciplinary Committee.
6. Moodle will be the resource sharing medium, keep checking the Moodle page and your emails regularly.

Tentative Course Outline

- See the document
  - Available on Moodle course page

Lets Start!

What is Data Mining and Why do we need it?

**Slides edited from Han and Kamber's online lecture slides*

Why Data Mining?

- The Explosive Growth of Data: from terabytes to petabytes
  - Data collection and data availability
    - Automated data collection tools, database systems, Web, computerized society
  - Major sources of abundant data
    - Business: Web, e-commerce, transactions, stocks, …
    - Science: Remote sensing, bioinformatics, scientific simulation, …
    - Society and everyone: news, digital cameras, YouTube
- <u>We are drowning in data, but starving for knowledge!</u>
- "Necessity is the mother of invention"—Data mining—Automated analysis of massive data sets

What Is Data Mining?

- Data mining (knowledge discovery from data)
  - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
  - Data mining: a misnomer?
- Alternative names
  - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Watch out: Is everything "data mining"?
  - Simple search and query processing
  - (Deductive) expert systems

Database Processing vs. Data Mining Processing

- Query
  - Well defined
  - SQL
- Output
  - – Precise
  - – Subset of database

- Query
  - Poorly defined
  - No precise query language
- Output
  - – Fuzzy
  - – Not a subset of database

Query Examples

- Database
  - Find all credit applicants with last name of Smith.
  - Identify customers who have purchased more than $10,000 in the last month.
  - Find all customers who have purchased milk
- Data Mining
  - Find all credit applicants who are low credit risks. (classification)
  - Identify customers with similar buying habits. (Clustering)
  - Find all items which are frequently purchased together. (association rules)

Data Mining Models and Tasks